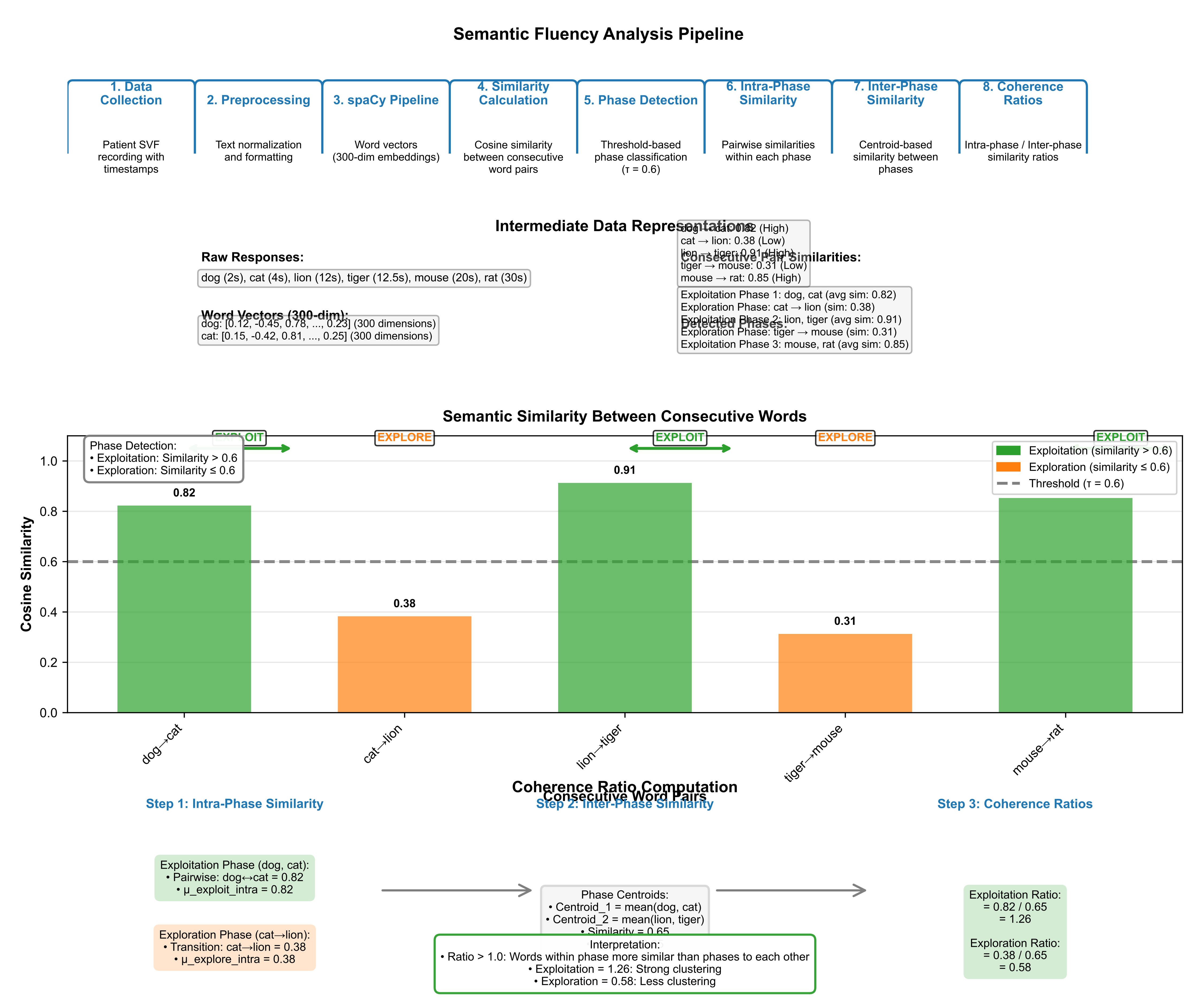
Semantic Fluency Analysis Pipeline
1. Data Collection
Patient SVF recording with timestamps
2. Preprocessing
Text normalization and formatting
3. spaCy Pipeline
Word vectors (300-dim embeddings)
4. Similarity Calculation
Cosine similarity between consecutive word pairs
5. Phase Detection
Threshold-based phase classification (τ = 0.6)
6. Intra-Phase Similarity
Pairwise similarities within each phase
7. Inter-Phase Similarity
Centroid-based similarity between phases
8. Coherence Ratios
Intra-phase / Inter-phase similarity ratios
Intermediate Data Representations
Raw Responses:
dog (2s), cat (4s), lion (12s), tiger (12.5s), mouse (20s), rat (30s)
Word Vectors (300-dim):
dog: [0.12, -0.45, 0.78, ..., 0.23] (300 dimensions)
cat: [0.15, -0.42, 0.81, ..., 0.25] (300 dimensions)
Consecutive Pair Similarities:
dog → cat: 0.82 (High)
cat → lion: 0.38 (Low)
lion → tiger: 0.91 (High)
tiger → mouse: 0.31 (Low)
mouse → rat: 0.85 (High)
Detected Phases:
Exploitation Phase 1: dog, cat (avg sim: 0.82)
Exploration Phase: cat → lion (sim: 0.38)
Exploitation Phase 2: lion, tiger (avg sim: 0.91)
Exploration Phase: tiger → mouse (sim: 0.31)
Exploitation Phase 3: mouse, rat (avg sim: 0.85)
Semantic Similarity Between Consecutive Words
Phase Detection:
• Exploitation: Similarity > 0.6
• Exploration: Similarity ≤ 0.6
EXPLOIT
EXPLORE
EXPLOIT
EXPLORE
EXPLOIT
Exploitation (similarity > 0.6)
Exploration (similarity ≤ 0.6)
Threshold (τ = 0.6)
0.82
0.38
0.91
0.31
Cosine Similarity
0.0
0.2
0.4
0.6
0.8
1.0
dog→cat
cat→lion
lion→tiger
tiger→mouse
mouse→rat
Consecutive Word Pairs
Coherence Ratio Computation
Step 1: Intra-Phase Similarity
Step 2: Inter-Phase Similarity
Step 3: Coherence Ratios
Exploitation Phase (dog, cat):
• Pairwise: dog↔cat = 0.82
• μ_exploit_intra = 0.82
Exploration Phase (cat→lion):
• Transition: cat→lion = 0.38
• μ_explore_intra = 0.38
Phase Centroids:
• Centroid_1 = mean(dog, cat)
• Centroid_2 = mean(lion, tiger)
• Similarity = 0.65
Interpretation:
• Ratio > 1.0: Words within phase more similar than phases to each other
• Exploitation = 1.26: Strong clustering
• Exploration = 0.58: Less clustering
Exploitation Ratio:
= 0.82 / 0.65
= 1.26
Exploration Ratio:
= 0.38 / 0.65
= 0.58